LINEAR REGRESSION

Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values.

It establishes a linear relationship between independent variables and a dependent variable using a best-fit line.

Types of linear regression

- Simple linear regression
- Multi linear regression

Why did you use train-test split?

To divide the dataset into:

- Training data (to train the model)
- Testing data (to evaluate performance)

This prevents overfitting and checks real-world performance.

Why did you use StandardScaler?

Because features have different ranges.
StandardScaler normalizes the data so that:

Z= x-mean/ standard deviation That

means:

Subtract the mean of the column. Divide by standard diviation of that column.

After scaling:

- Mean = 0
- Standard deviation = 1

This improves model stability and performance.

When to use Standard Scalar

Works well for algorithms that depend on distances or gradients, such as:

- Linear Regression
- Logistic Regression
- K-Nearest Neighbour (KNN)
- Support Vector Machines (SVM)
- Principal Component Analysis (PCA)
- Neural Network (NN)

What is the purpose of the correlation heatmap?

The heatmap shows the relationship between features. It helps:

- Identify strongly correlated features
- Detect multicollinearity

What is the meaning of intercept?

Intercept is the predicted value of the target variable when all independent variables are zero.

What do coefficients represent?

Coefficients show the impact of each feature on the target variable.

Positive coefficient → increases house value

Negative coefficient → decreases house value

Mean Squared Error (MSE)

MSE measures the average squared difference between actual and predicted values.
Lower MSE indicates better performance.
It penalizes large errors more than small ones.

R^2 Score

R^2 Score measures how well the model explains the variance in the data.

Range:

- 0 → Poor model
- 1 → Perfect model

Higher R^2 means better model fit.

Why did you plot Actual vs Predicted values

To visually check model accuracy.
If points lie close to the diagonal line, predictions are good.

What are the assumptions of Linear Regression

1. Linearity
2. Independence
3. No Multicollinearity
4. Homoscedasticity
5. Normal distribution of errors